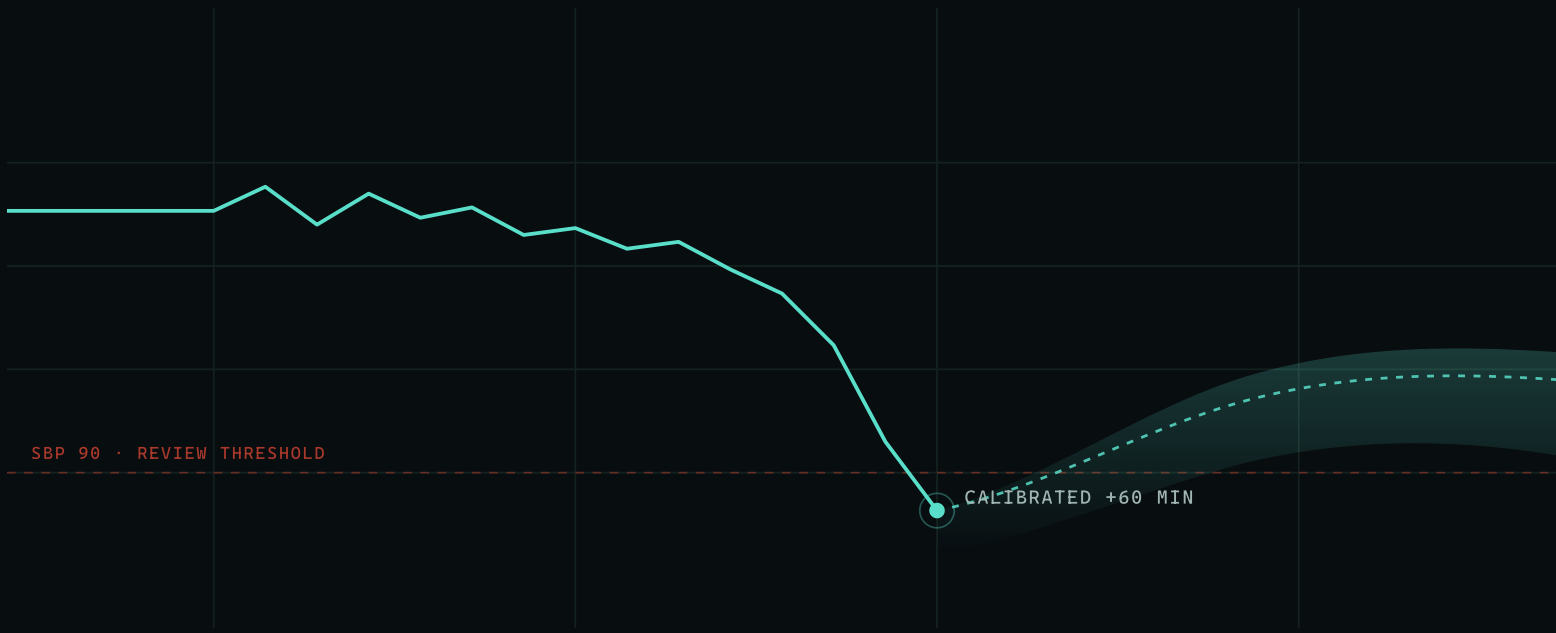
H HEMOTWIN V2 · RESEARCH

See the drop before it lands.

What the models actually do, what the evidence numbers mean, and what this system can honestly claim before it reaches a chair.



RESEARCH PROTOTYPE · NOT A MEDICAL DEVICE · DISPLAY-ONLY · NO PRESCRIPTION

205,482 SESSIONS · 1,070 PATIENTS · FOLD 2 UNTOUCHED

THE HONEST SUMMARY

A calibrated forecast, exact arithmetic, and a review pathway — not a treatment engine.

HemoTwin mirrors a dialysis session, projects the next hour of systolic blood pressure, and structures what the unit should review. It does not decide treatment. The what-if UF effect was tested and not found, so it stays clearly labelled as an illustration.

FORECAST ERROR
10.99
mmHg typical miss (MAE), fair 59-feature model, fold 2.

CHAIR-RISK RANKING
0.831
AUROC for pre-session crash risk (A2), fold 2.

UF EFFECT TEST
+0.13
mmHg per 0.25 L/h; 95% CI includes zero. Verdict: fail.

ALERT AUTHORITY
0
The event model is shadow-only; it does not alarm or page.

WHAT WE CAN CLAIM

- A calibrated pre-anchor forecast of the 60-minute cuff-sampled nadir, with a 90% band.
- Exact volumetric fluid arithmetic, including what a scenario leaves behind for the next session.
- A deterministic, versioned review pathway and a local, auditable session record.
- Honest negative results: the what-if twin and the oracle comparison are reported as failures.

WHAT WE REFUSE TO CLAIM

- That reducing UF early reliably raises SBP, or that any option is minimum-effective.
- Validated alert utility, diagnosis, prescription, or autonomous machine action.
- A treatment effect from the mechanism animation, or a per-minute confidence band.
- External, prospective, symptomatic-IDH, or Indian-unit validation.

How to read this brief

Every number is labelled with the model and the data split it belongs to, because several numbers in this project look similar but are not interchangeable. Each section pairs a figure with a plain-language reading: *what it means*, and *what it does not*.

01 The two models — forecast + chair risk
02 Evidence tab, decoded — G1–G7
03 Why this matters — staffing context
04 System diagram — what runs where
05 The alarm budget — attention as a resource
06 The fluid math — worked example
07 Why interventions break it — and how to fix it
08 Into the workflow — what nurses see
09 Next direction — wearables & evidence
– Sources & boundary — references

THE ONE-SENTENCE VERSION

The forecast is useful as a projection, the arithmetic is trustworthy as arithmetic, and the review pathway is trustworthy as process. The treatment effect is not established — so the system shows the trade-off instead of pretending to know it.

THE ORACLE TRAP, STATED ONCE

The 9.60 mmHg display forecast consumes 13 observed *future* UF features. It looks better because it already knows what the machine will do. It is a display-only card, not the serving model, and never a like-for-like comparison.

1A · CASE LAB FORECAST

A 60-minute nadir forecast, not a treatment oracle.

The curve in Case lab is a LightGBM gradient-boosted model. It reads the session up to a blood-pressure cuff and predicts the lowest SBP the cuffs will record in the next hour. The solid line is the calibrated projection; the band is its 90% nadir interval.

Model at a glance

TASK	Predict the 60-minute cuff-sampled nadir from the current anchor.
MODEL	LightGBM GBDT, 3-seed ensemble, nadir-specialised refit (Forecast-v2 lineage).
SERVING CONTRACT	59 pre-anchor features; no future UF; isotonic calibration; 90% band.
TRAINING DATA	HEMOBP, Taiwan: 1,070 patients, 205,482 sessions; fold 2 untouched (12,651 anchors).
ROLE	Display-only projection. It never gates the review pathway.

What each score means

SCORE	VALUE	PLAIN READING
MAE	10.99 mmHg	A typical miss is about 11 mmHg. At a 90 threshold, that is the gap between 90 and 79 or 101 — which is why the band matters.
RMSE	14.65	Higher than MAE, so some sessions miss badly. The tail is real.
Bias	−2.42 mmHg	Predictions sit slightly *below* the observed nadir — cautious, not a safety margin.
Band coverage	89.2% / 90.1%	The nominal 90% band contains the observed nadir about nine times in ten (HOLD / TAPER).
G1 taper MAE	11.60 vs 15.55	On 374 taper anchors the new surface is 25% closer than the retired v3 twin.

What the model sees

59 causal features, all built from data available at the anchor. The 13 future-UF features are deliberately excluded; a separate 72-feature *oracle display* reaches 9.60 mmHg but is not deployable and is not the comparison baseline.

FAMILY	REPRESENTATIVE FEATURES	WHY IT IS THERE
Blood pressure	`sbp_now, dbp_now, map_now; means, minima and slopes over 15 / 30 / 60 / 120 min; drop from session max; pulse pressure; SBP variability`	The current level and the shape of the fall.
Ultrafiltration	`uf_Lh, cum_uf_L, uf_ml_kg_h, frac_uf_done, uf_slope_15/30, uf_over13`	How hard fluid is being pulled now, and how much is left.
Machine	`blood_flow, conductivity, dia_temp, temp_dev, cond_dev`	Settings that change the physiological response.
Session context	`dry_weight, pre_weight, elapsed_min, gap_since_last, n_bp_60, month`	Position in the session and measurement cadence.
Twin state	`k_pat, beta_pat, twin_D_now, twin_refill_Lh, twin_refill_ratio, twin_pred_nadir`	A compact physiological state description; it is an input, not a validated effect model.
Patient history	`hist_idh_rate10, hist_mean_nadir3/10, hist_prev_nadir, hist_early_crash10, hist_last_idh`	Strictly earlier sessions only; prior crashes and prior nadirs.
Static	`age, sex_M, dm, vintage_years`	Baseline risk; vintage carries real signal.

WHAT THE FORECAST MEANS IN PRACTICE

Use it as a **direction and a range**: "the next hour is likely to land near 86, with a band from 67 to 105." It is not a per-minute confidence band, and it is not a promise about symptoms.

Sources: model card and FACTS [2]; forecast-v2 feature contract and fold-2 confirm [3]. Fold 2 = patients never seen in training or tuning; its result was read once.

1B · CHAIR BOARD RISK (VARIANT A2)

A pre-session risk score that is computable before the first cuff.

The Chair board answers a different question from the forecast: *at session start, which chairs deserve a closer look?* A2 reads history, static patient fields and the planned UF goal — no blood pressure and no current UF rate — so it cannot leak the outcome it is trying to predict.

Model at a glance

TASK	Observational pre-session risk of a crash (SBP < 90) during the session.
MODEL	LightGBM booster on a 31-column surface including missing-value indicators; variant A2.
FEATURES	Prior IDH rate and nadirs, early-crash share, interdialytic interval and weight gain, planned UF goal, age, sex, diabetes, vintage, dry weight. Excludes `sbp_anchor` and anchor-time `uf_ml_kg_h`.
SPLIT	Fold 2: 12,651 sessions, 3,109 crashes (24.6%); patients never in training.
ROLE	Attention cue. Fixed chair order; never an allocation or priority order.

What each score means

SCORE	VALUE	PLAIN READING
AUROC	0.8313	Pick a crashed and a non-crashed session at random: the crashed one ranks higher 83% of the time.
Recall@25	0.6015	Review the top 25% of chairs and you capture 60% of the crashes — 40% are outside the referral set.
PPV@25	0.5914	Of the chairs in that top quarter, 59% actually crashed; 41% did not.
ECE10	0.0184	Calibration error about 1.8 probability points: predicted percentages are close to observed rates.

What the bands actually look like on fold 2

BAND	CUT	SHARE OF SESSIONS	CRASH RATE INSIDE	HOW TO READ IT
HIGH	≥ 0.35	27.8%	57.0%	Most of these crash, but 43% do not. Not a diagnosis.
MODERATE	0.20–0.35	12.7%	27.1%	A mixed group; useful only as a review prompt.
LOW	< 0.20	59.5%	8.9%	Lower risk, not safe: 21.5% of all crashes were scored low.

THE SENTENCE THAT MUST TRAVEL WITH THE SCORE

A low band is not safe, and a high band is not a diagnosis. In the curated five-chair demo, 1 high chair and 0 low chairs crashed — that is a story chosen to show the range, not an accuracy estimate.

WHY A2 AND NOT THE BETTER-LOOKING MODEL

The anchor-time models score higher (B: AUROC 0.8788). They also read BP and UF from roughly 90 minutes into the session, which makes them partly circular as a *pre-session* card. We accept the lower A2 score so the number is available before the first cuff.

What this changes for a unit

At a 25% referral burden (two of eight chairs), the board points attention at roughly 1.2 true crashes for every two chairs reviewed — while missing about 40% of crashes. That is why it is a **review-attention surface**, not a triage rule, and why the board order stays fixed by chair number. The value is in starting the conversation earlier, not in certifying anyone as safe.

Feature honesty: the planned UF goal encodes clinician behaviour, not physiology, and 391 first sessions without history are imputed from stored medians and flagged. A2 is observational — it can say "sessions that looked like this often crashed," never "lowering this goal would prevent a crash." Sources: CRASHRISK [3]; crashrisk pre-registration; evidence/chair_risk.json. Recall@25 and PPV@25 use the pre-registered top-25% referral set.

WHAT THE NUMBERS IN EVIDENCE ACTUALLY MEAN

Seven pre-registered checks — not a score, and not clinical validation.

The Evidence tab is a transparency page. "Pass" means an engineering criterion written down before the result was seen was met. It does not mean a patient benefit was demonstrated. The tab's most important finding is a *failure*: the UF what-if test.

G1 PASS · G2 PARTIAL · G3 FAIL · G4 PASS · G5 PASS · G6 PASS · G7 FAIL · 5 OF 6 GATE BOOLEANS TRUE

WHAT YOU SEE	THE NUMBER	PLAIN LANGUAGE	WHAT IT IMPLIES
G1 · taper	MAE 11.595 vs 15.550 ratio 0.746	On sessions where UF was tapered, the new pre-anchor surface predicts the nadir with 25% less error than the retired v3 twin.	The new substrate is better than the old one on the slice that matters for tapering. It does not say the taper helps.
G2 · baseline	10.99 / 9.60 fair / oracle	The fair model ties its own base at gain 0 and loses to the oracle display forecast, which uses 13 future-UF features.	We pass against our own baseline, fail against an oracle we cannot deploy. Both are reported.
G3 · actual vs hold	ΔMAE 0.000 CI [0, 0]	At the calibrated gain of 0, the "actual UF" and "hold UF" projections are identical.	The what-if mechanism was not validated. This is the reason the mechanism sketch is off by default.
G4 · monotonicity	0 violations 3,086,844 checks	Stepping UF down more never produces a worse projected pressure at unit gain, and the projection is continuous at the anchor.	The simulation is internally coherent — an engineering pass, not evidence of a real effect.
G5 · band coverage	89.2% HOLD 90.1% TAPER	The 90% uncertainty band contains the observed nadir about 90% of the time on held-out sessions.	The uncertainty is roughly honest at the population level. It is marginal coverage, not a per-patient guarantee.
G6 · invariance	200/200 284 cold starts	Changing future UF records does not change the pre-anchor prediction, and new patients stay bounded.	No future leakage; the score is computable at the moment it claims.
G7 · UF effect	+0.13 mmHg CI [−2.63, +3.06]	Reducing UF by 0.25 L/h early raised SBP by 0.13 mmHg more than matched controls — indistinguishable from zero.	**Verdict: fail.** The twin remains a labelled mechanism illustration; operational gain 0.

HOW TO READ A FOREST PLOT

The dot is the estimated effect; the horizontal line is its 95% confidence interval. If the interval crosses zero, the data cannot distinguish the effect from no effect. G7's interval runs from −2.63 to +3.06 mmHg, so "no detectable effect" is the honest reading — not proof that UF never matters.

Sham control +2.91 mmHg passed its own pre-trend but failed as a placebo check; **P6a +0.43 mmHg** means the out-of-sample trajectory error got slightly worse, not better.

THE OTHER ROWS IN THE TAB

Temporal-order diagnostic: of 397 actual-vs-hold rows, 234 UF cuts happened at the anchor with no preceding observed drop. That weakens the earlier "staff reacted to a fall" explanation, but it does not create a positive effect.

Protocol hash, FHIR bundle, audit chain: engineering evidence that the pathway and record are reproducible and exportable. They are *not* clinical evidence, and the hash chain is a local demo object, not an immutable clinical record.

THE ONE LINE TO TAKE AWAY

"5 of 6 pass" is an engineering scoreboard. The clinically decisive result is that the pre-registered UF effect test **failed**. A product that hides that failure would be less trustworthy than one that prints it.

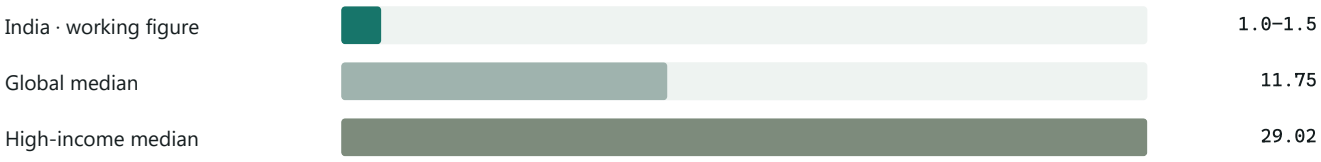
Sources: EVIDENCE_STATUS [3]; evidence/g7_report.json; demo_twin.json; crashrisk confirm files. G2 is a vacuous self-tie at gain 0 and is marked partial for that reason.

THE STAFFING GAP BEHIND THE PRODUCT

The person at the chair often has the least support and the most immediate decisions.

Haemodialysis is a high-frequency, high-consequence workflow: roughly three sessions per patient per week, with a doctor usually covering many chairs and sometimes several units. The gap is not another risk score. It is time, shared language, and explicit trade-offs.

NEPHROLOGIST DENSITY · PER MILLION POPULATION



Bars are relative to 29.02 pmp. India's 1.0–1.5 is a project working figure consistent with ISN-GKHA country data; global and income-group figures are from ISN-GKHA 2023. [4][5]

INDIA · WORKING FIGURE

1–1.5

nephrologists per million — roughly one per 0.7–1 million people; about 20× below high-income density. [2][4]

UTTAR PRADESH EXAMPLE

145

nephrologists in a state of ~240 million. Only three centres provide dialysis-technician courses. [7]

PMNDP FOOTPRINT

13,824

haemodialysis machines across 1,925 centres in 751 districts (official dashboard). [6]

The scenario the system is built for

In many district hospitals under the national dialysis programme, the chair is run by nurses or technicians with limited formal nephrology training; the nephrologist is remote and covers a large population. The Uttar Pradesh review is explicit: most dialysis at district hospitals is performed by nurses or technicians with limited working knowledge, and technician-course capacity is scarce. [7] This is not a criticism of the staff — it is a description of the support they are given.

A technician who learned from experience can recognise "this patient looks off," but has little time to work through the fluid arithmetic, no shared vocabulary for a remote doctor, and no structured list of what to check before calling. The result is predictable: calls happen late, handoffs are verbal, and the same trade-off is rediscovered at every chair.

WHY A SUPPORT LAYER CAN MAKE A DIFFERENCE

Not because it predicts better than a person. Because it **mirrors the session** (what has actually happened), **structures the review** (what to check, in what order), and **shows the arithmetic** (what a choice leaves behind). That buys the technician time and gives the remote nephrologist a shared, compact picture before the call.

THE SCALE WE ARE TALKING ABOUT

HEMOBP records a hypotensive nadir (<90 mmHg) in **10.4%** of sessions. The Spanish cohort records clinician-defined symptomatic hypotension in **25.5%** — different labels, never combined. Applying the 10.4% rate to an estimated ~34 million Indian sessions per year gives roughly **3.5 million** hypotensive episodes a year. That is arithmetic on estimates, not a measured burden. [2][3]

AND THE STAFF ALREADY ACT

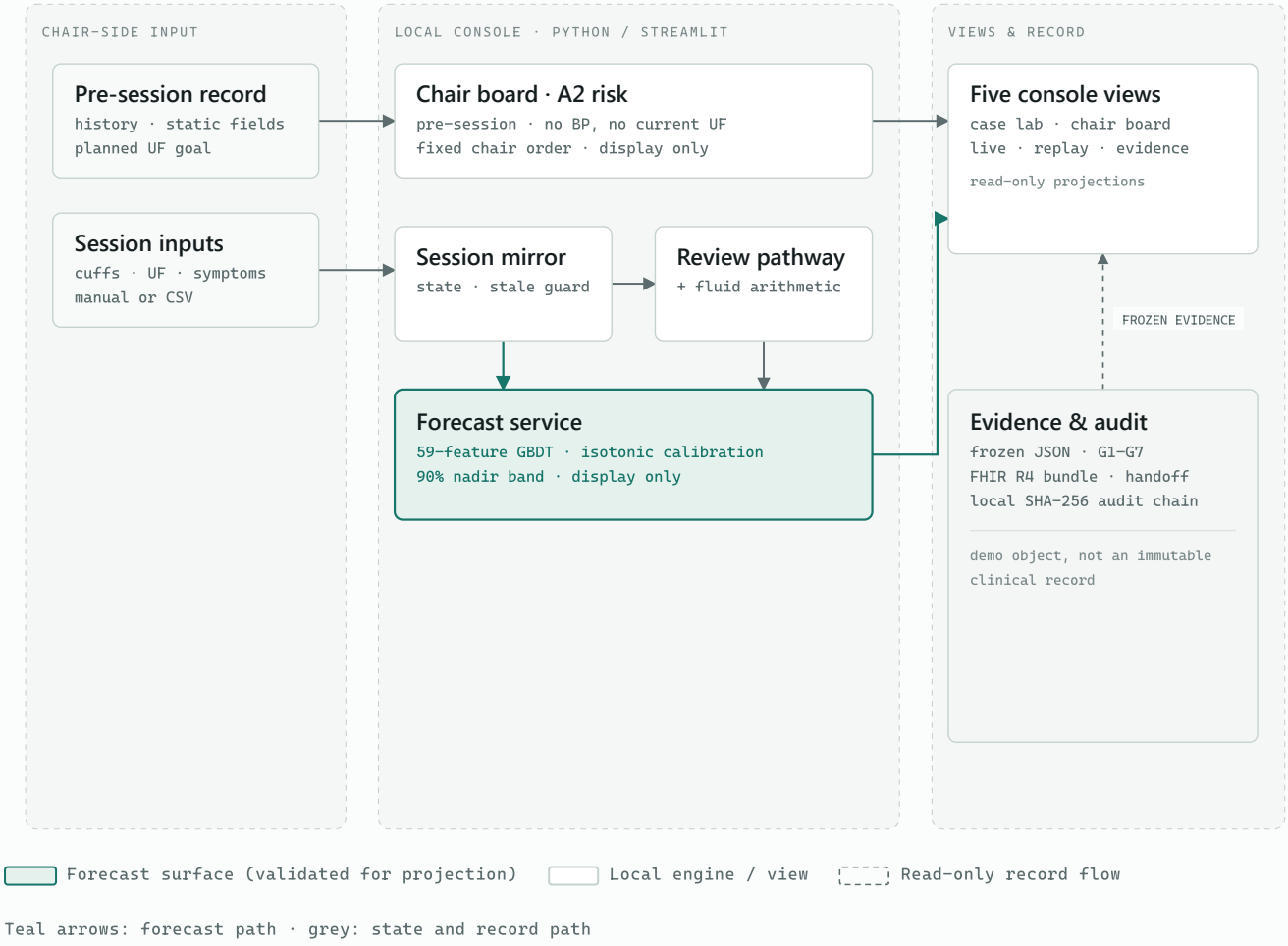
In HEMOBP, staff cut UF by more than 20% mid-session in **87.2%** of sessions; the median rate falls from **0.58 to 0.10 L/h**. The intervention is not missing — what is missing is a shared record of *why*, *when*, and *what it cost*. That is exactly the layer HemoTwin tries to add. [2]

Sources: ISN-GKHA 2023 / Lancet Global Health 2024 [4]; ISN Atlas income-group figures [5]; PMNDP official dashboard [6]; Indian J Nephrol, Uttar Pradesh services review [7]; HemoTwin FACTS [2]. The India density figure is a planning figure, not a census.

WHAT RUNS WHERE

One local console, one record — no network call at runtime.

The bay is a separate WebGL build. The console is Python/Streamlit. Models, protocol rules and fluid math run locally; the only outbound interface is an export the user triggers.



NO HIDDEN CLOUD

The app makes no network call at runtime. Models, evidence and audit are files on the machine; exports are user-triggered.

DEVICE-AGNOSTIC BY DESIGN

Cuffs, UF and symptoms are entered manually or imported from CSV. Machine integration is a future adapter, not a dependency.

THE HUMAN STAYS IN CHARGE

The pathway is a checklist, not an order. The forecast is a shadow; measured rules set the review state.

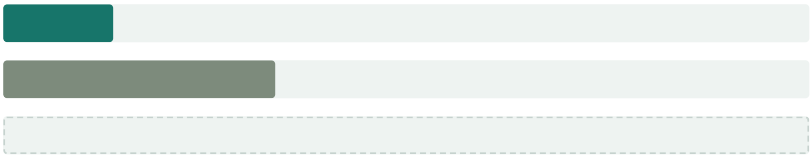
The what-if ODE twin is intentionally not on the validated path in this diagram; it appears only as a labelled illustration inside Case lab. Evidence artifacts are frozen JSON generated from the research workspace.

ATTENTION IS THE SCARCE RESOURCE

Why a model with AUROC 0.88 still lost its alert authority.

An alarm budget is the number of false or non-useful alerts per session a unit is willing to absorb before staff start ignoring the system. A model can rank well and still fail at a fixed budget — because ranking is not the same as an operating point.

TIMELY DETECTION AT THE PRE-REGISTERED BUDGET (≤0.25 NON-USEFUL ALERTS PER SESSION)



The gate was written before the result. The model missed it by a wide margin and lost alert authority. [3]

What the numbers say

QUANTITY	VALUE	MEANING
AUROC (Task 02)	0.880	Good ranking — but ranking alone does not save a fixed alert budget.
Detection at 0.25/session	8.2%	Less than one in twelve crash sessions was caught in time.
Rescue rule at 0.26	14.1%	A simple SBP + refractory rule caught roughly twice as many on fold 2; the model score added burden, not alerts. [3]
Sensitivity @ 90% specificity	0.721	A different operating point: fewer false alarms, but about one third of events missed.
Shipped historical point	18.2%	False-alarm session rate — roughly one false alarm every five sessions. Retired.

What it means on a ward

At eight chairs, 0.25 alerts per session is about **two extra alerts per treatment round**; an 18.2% false-alarm session rate is about **1.5 false alarms per round**. That is the real cost: not the model's error, but the nurse's attention.

The event model is therefore **shadow-only**: it never pages, ranks, or gates the review pathway. The working alert is a measured rule (SBP below ~99 with a non-resetting 60-minute refractory), and even that is a retrospective replay result, not a prospective safety claim.

> THE TRADE-OFF IN ONE SENTENCE
>
> Lower the threshold and you catch more crashes but drown the unit in alarms; raise it and you miss the ones that matter. The product exposes the threshold instead of hiding it.

The Chair board has a budget too: 25% referral

The pre-registered referral rule is "review the top 25% by risk" — at eight chairs, two chairs. On fold 2 that captures **60.15%** of crashes and **59.14%** of referred chairs crash. It misses about **40%** of crashes and mislabels about **41%** of referrals. That is why the board is fixed-order, display-only, and shifts attention rather than replacing it.

2 OF 8 CHAIRS · ~1.2 TRUE CRASHES PER ROUND · ~40% OF CRASHES OUTSIDE · 21.5% OF CRASHES SCORED LOW

Sources: Task 02 alert-utility evaluation; alert-v1 rescue-rule sweep; FACTS [2][3]. "Non-useful alerts per session" counts alerts that did not precede an event in the warning window. Retrospective replay only; no prospective alert claim.

EXACT ARITHMETIC, NOT A PHYSIOLOGICAL PREDICTION

Where the fluid goes when a session ends early.

The arithmetic answers a narrow, honest question: if this much UF is left unfinished, and this much extra fluid is given, how much is carried into the next session — and what does that do to the prescribed intensity? It does not predict SBP.

The five formulas

```
# goal, removed and carry are litres (converted to mL)
remaining = max(goal − removed, 0)
left = q_left × remaining
carry = left + extra_fluid
next_volume = next_prescription + carry

# reference flag, not a prescription limit
intensity = next_volume_mL / (weight_kg × duration_h)
flag = intensity > 13 mL/kg/h
```

q_left is the fraction of the remaining prescription the scenario does not deliver (0 = finish, 1 = stop). **extra_fluid** is an added bolus or intake assumed to stay in the patient until session end. The 13 mL/kg/h line is the KDOQI reference ceiling used as a flag — not an order.

Worked example

INPUT	VALUE
UF goal	3.00 L
Already removed	1.20 L
q_left (unfinished fraction)	0.50
Extra fluid (bolus)	0.25 L
Next prescription · duration · weight	3.00 L · 4 h · 70 kg

```
remaining = 3.00 − 1.20 = 1.80 L
left = 0.50 × 1.80 = 0.90 L
carry = 0.90 + 0.25 = 1.15 L
next_volume = 3.00 + 1.15 = 4.15 L
intensity = 4150 / (70 × 4) = 14.8 mL/kg/h → over 13
```

READING THE WORKED EXAMPLE

Half-finishing the UF and giving a 250 mL bolus leaves 1.15 L to carry forward. The next session's prescription is unchanged at 3.00 L, so its true starting load is 4.15 L — about 14.8 mL/kg/h, above the 13 reference. The arithmetic makes that cost visible; it does not say whether accepting it is clinically right.

The projection model (labelled illustration)

Case lab also draws a dashed mechanism sketch. It is an ODE with two fitted patient parameters and a literature-anchored immediate-refill term; the calibrated gain is **0**, so it does not move the solid forecast. It is shown to make the physiology legible, not to claim an effect.

```
dD/dt = (1 − ϕ) · UF/60 − k · D # plasma-deficit D, UF in L/h
ϕ = 0.30 immediate refill fraction (Wang 2023 range 30–50%)
k ≈ 0.245 /h population refill rate (session IQR 0.10–1.31)
β ≈ 16.3 mmHg/L population SBP sensitivity to deficit
```

WHAT EACH SYMBOL MEANS

φ: share of each litre removed that is replaced immediately by refill. **k**: how fast the remaining deficit refills. **β**: how many mmHg the pressure falls per litre of deficit. **Gain**: how strongly the simulation lets a UF change move SBP; it calibrated to 0.

WHY IT IS NOT A TREATMENT CLAIM

The what-if test (G7) found **+0.13 mmHg per 0.25 L/h**, CI **[−2.63, +3.06]** — no detectable effect. At gain 0, actual and hold projections are identical (G3). The animation is honest only while it stays labelled as a mechanism illustration.

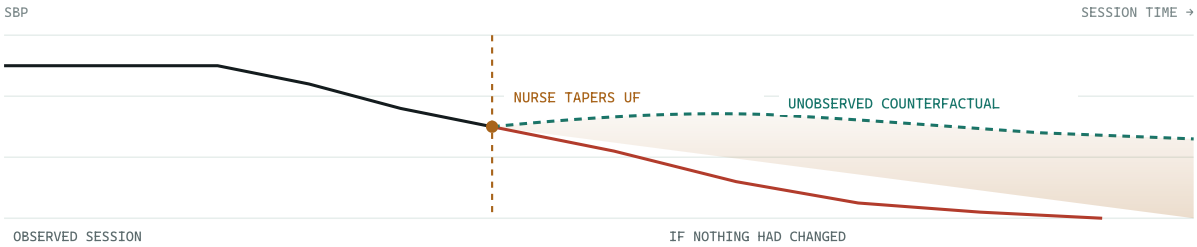
Assumptions shipped with the arithmetic: additional fluid stays in the patient; carry-forward is added only in the displayed scenario; 13 mL/kg/h is a reference flag. Sources: src/session_twin/fluid.py; KDOQI 2015 [10]; Wang 2023 / Mitsides 2019 [9]; G7 report [12].

THE HARDEST PROBLEM IN THE DATA

The nurse changes UF. The future changes too.

A forecast learns from observed sessions. In those sessions, staff already respond to falling pressure: they taper, pause, or give fluid. That makes the treatment and the outcome entangled — and it is why a model trained on the record cannot answer "what would have happened if we had acted differently?"

CONCEPTUAL TIMELINE — ILLUSTRATION, NOT DATA



Why the simple version fails

1. Confounding by indication. Staff taper when the patient is already deteriorating. In the record, "UF was reduced" often precedes a crash — not because the taper caused it, but because it was a response to the fall. A model that learns that association learns the opposite of the causal lever.

2. The action is not known at prediction time. At the anchor, the future taper has not happened. A feature like "UF changed by −0.3 L/h at minute 95" cannot be used by a forecast made at minute 60 without leaking the future.

3. The policy changes when you deploy. If the console changes what staff do, the next data distribution changes too. A model trained on yesterday's practice is calibrated to yesterday's decisions.

4. The measurement is too sparse. Cuffs arrive roughly every 30 minutes. The immediate response to a UF change can wash out between readings, and the labels inherit that cadence.

5. The effect may be genuinely small. The G7 experiment estimated **+0.13 mmHg per 0.25 L/h** with a CI from −2.63 to +3.06 — indistinguishable from zero. That does not prove UF never matters; it does mean the data cannot support a confident positive claim at this scale.

THE PRACTICAL CONSEQUENCE

The what-if sketch is off by default. The console shows arithmetic (what a choice leaves behind) and a projection under the observed trajectory — not a promise about what an intervention will do.

THE DIFFERENCE IN ONE LINE

What the model learns: P(crash | a taper happened). **What a decision needs:** P(crash | we choose a taper). Those two are equal only if tapers were assigned at random — and they were not: staff tapered because the patient was already falling. That is confounding by indication, and no extra feature removes it without a change in design.

Supporting facts: staff cut UF >20% mid-session in 87.2% of sessions; median 0.58 → 0.10 L/h; 81.7% end below 80% of peak UF [2]. G7: 581 matched events, 158 patients, MDE 2.33 mmHg; sham +2.91; P6a +0.43 [3][12].

SO CAN WE JUST ADD “UF CHANGED ±X” AS A FEATURE?

Yes for nowcasting. No for causality.

Action history is useful — it is one of the strongest signals that a clinician has seen something the cuffs have not yet shown. But it answers a different question from “what will this action do?” The distinction decides what you are allowed to ship.

USE OF ACTION DATA	WHAT IT CAN IMPROVE	WHAT IT CANNOT ANSWER	RISK
Past changes `ΔUF, time since change, cumulative change, reason flag`	Short-horizon nowcasting: the action is a proxy for the clinician’s assessment, so the next cuff becomes easier to anticipate.	The effect of a *new* action, or what would have happened without the past one.	Can encode clinician behaviour and drift when practice changes.
Future changes `next UF segment, future rate`	Display metrics — that is the 9.60 oracle card.	Anything usable at decision time; it already knows the future.	Leakage: better numbers, no deployable predictor.
Joint action + outcome model `propensity / policy head`	Forecasts conditioned on a planned action, with inverse-probability weighting; supports “what if the unit follows this protocol?”	A causal effect without assumptions about unmeasured confounding.	Fragile under policy shift; needs action logs and overlap checks.
Designed experiment `target trial emulation, protocolised step-down`	An actual effect estimate with a pre-registered estimand and power calculation.	Quick answers — effects are small and need many matched events.	Ethics, cost, and a real prospective protocol.

The route we would take next

1 · Log the action, not just the state. Every UF change gets a timestamp, a rate delta, a duration and an optional reason. This is the dataset the current record lacks — and it is a prerequisite for every option above.

2 · Separate the two products. Keep the state mirror and the observational forecast honest; build any effect estimator as a separate, clearly labelled research surface with its own validation.

3 · Start with nowcasting. Test whether past-action features improve next-cuff prediction under a patient-disjoint split. If they do, ship them as a prediction feature — never as a treatment recommendation.

4 · Then identify an effect. Use target-trial emulation with time-varying confounders, or a protocolised/randomised step-down with a pre-registered SBP endpoint. G7 shows the effect is small enough that the design, not the model, is the limiting factor (MDE 2.33 mmHg).

5 · Increase the measurement density. Continuous or 5-minute BP, plus symptom and action labels, would let the data see a response that 30-minute cuffs cannot resolve.

> THE DISCIPLINE
>
> Never let a prediction model masquerade as a treatment effect. The current console enforces that separation in its copy, its gates, and its default view.

WHAT “ADDING THE FEATURE” WOULD ACTUALLY DO

If you add “UF changed” to a model trained on historical sessions, the model will likely learn *the clinician’s policy*: tapers happen when pressure falls, so the feature is a proxy for deterioration. The model may predict the next cuff better and still be useless as a guide to action. That is not a bug in the feature; it is the difference between **prediction** and **intervention**. The only way to close that gap is a design that makes the action exogenous — a protocol, a randomised step, or a defensible natural experiment — plus a pre-registered effect estimate.

Directions are research plans, not current capabilities. See EVIDENCE_STATUS “Next evidence required” [3].

WHAT THE NURSE ACTUALLY SEES

A five-step loop that fits the chair, not a new workflow.

The console is designed around the moments that already exist: before the first cuff, at each cuff, when something looks wrong, and at handoff. It adds structure to those moments without taking the decision away from the team.

01 · PRE-SESSION
Look at the board
Fixed-order chairs with an A2 risk band. No BP, no current UF. Attention cue only.

02 · START
Enter the session
Starting BP, UF goal, dry weight, symptoms. The mirror begins; missing data is flagged.

03 · EACH CUFF
Read the projection
Nadir forecast, 90% band, protocol state, stale-cuff guard. The forecast never sets the state.

04 · IF REVIEW
Work the pathway
Assess, repeat BP, review UF, check fluid. Role-scoped steps, deterministic order.

05 · HANDOFF
Export the record
FHIR bundle, bilingual card, frozen evidence, local audit chain. Replay for teaching.

What changes for the technician at the chair

MOMENT	WITHOUT THE CONSOLE	WITH THE CONSOLE
Before the cuff	A list of names and a planned UF goal.	A fixed board with a pre-session risk band and the planned goal.
Falling pressure	A number and a judgement call.	A 60-minute projection with a range and the measured state.
Calling the doctor	A verbal summary and a guess at the fluid balance.	A compact handoff with the same numbers and the arithmetic.
Change of plan	Decision made; the cost is remembered, not recorded.	Carry-forward arithmetic is explicit; the action is logged.

What it helps with. Time: the projection and the checklist shorten the “what do I do now?” pause. Shared language: the same five views travel with the patient from the chair to the phone call. Memory: the replay and audit chain preserve what was seen and done.

What it does not do. It does not alarm, prescribe, allocate chairs, or act on the machine. The protocol state is driven by measured rules (SBP, symptoms, drop from baseline), not by the forecast. Missing required data produces *INSUFFICIENT_DATA*, not a guess.

INTEGRATION PATH

Today: manual entry or CSV on a local machine, with no machine connection and no network call. Next: a read-only machine-log adapter and a local deployment package. Later: identity, role enforcement, and a validated interface — all require a pilot and ethics approval first.

Who does what

TECHNICIAN / DIALYSIS NURSE
Enters cuffs and UF; reads the band and the state; works the pathway; escalates. The primary user at the chair.

REMOTE NEPHROLOGIST
Receives the handoff; sees the same numbers; decides. The console is a shared picture, not a second opinion engine.

UNIT LEAD / CLINICAL ENGINEER
Owns the protocol version, the local calibration, the audit review, and the override log for any pilot.

The current prototype has no authentication, no role enforcement, no machine integration, and no external notification. The audit chain is a local demo object. These are prototype limits, not hidden features.

WHAT WOULD MAKE THIS REAL

Earn the right to predict, then earn the right to advise.

The next phase is not a bigger model. It is better evidence, denser measurement, and a clear separation between a mirror, a forecast, and a treatment effect.

NOW · 0–6 MONTHS

- Local Indian cohort + label harmonisation (SBP<90 vs symptomatic).
- Log every UF change with time, delta, duration, reason.
- Prospective silent-mode run: does the workflow fit the chair?
- Recalibrate on the local unit; publish subgroup metrics.

NEXT · 6–18 MONTHS

- Action-aware nowcasting, evaluated as prediction only.
- Pre-registered causal pilot (target trial or protocolised step-down).
- Continuous / 5-minute BP feasibility and symptom labels.

LATER · 18 MONTHS+

- Wearable fusion — only after incremental-value studies.
- Read-only machine integration and local deployment package.
- Regulatory strategy (CDSCO SaMD, DPDP Act 2023).
- Prospective outcome evaluation, not a model demo.

Could wearables add signal?

Yes — as research features, not as a cuff replacement. Dialysis is a 3–5 hour window with repeated sitting still, which is unusually friendly to optical and motion sensing. The constraint is not the sensor; it is validation, missingness, and workflow.

SENSOR	SIGNAL	POTENTIAL FEATURE	CAVEAT
Wrist PPG / chest strap	Heart rate, HRV	Beat-to-beat response to falling volume; early sympathetic activation	Motion artefact; hypotension can also cause bradycardia; needs per-patient baselines.
Pulse oximeter	SpO_2, pulse	Peripheral perfusion during UF; desaturation episodes	Finger probes already in some units; not a BP surrogate.
Accelerometer	Movement, posture	Restlessness, sleep, still periods; artefact detection	Indirect; dialysis chairs constrain movement.
Skin temperature / EDA	Thermoregulation, stress	Cooling and vasoconstriction patterns around events	Noisy; ambient temperature confounds.
Cuffless BP (research)	PPG morphology	Continuous trend between cuffs	Not validated for dialysis; cannot replace the cuff today.
Bioimpedance	Fluid status	Refill and volume state	Sensitive to placement and posture; not a drop-in wearable.

WHAT A CREDIBLE WEARABLE STUDY WOULD REQUIRE

Clock synchronisation to the cuff and machine; a prespecified incremental-value metric (does it improve MAE or recall at the same alarm budget?); patient-disjoint validation; handling of missing and artefactual windows; and a clear separation of monitoring from alerting.

WHAT WOULD NOT BE CREDIBLE

A wearable demo that correlates with BP in a convenience sample. The bar is incremental value over the measured cuff + UF state, with the same clinical claim boundary.

THE PRODUCT THESIS, UNCHANGED

The gap is time and a shared trade-off. A mirror, a structured review, and honest arithmetic buy the technician enough time to react and seek advice. Prediction is one feature of that — not the product.

REFERENCES

Where every number comes from.

Internal numbers are from the frozen v2 evidence package (11 Nov 2026). External context is cited below. Where a figure is a planning estimate, it says so.

Internal evidence

- **[2]** HemoTwin v2 model card (2026-11-10) and `docs/FACTS.md`: forecast lineage, features, scores, India context, published-work comparisons.
- **[3]** HemoTwin `docs/EVIDENCE_STATUS.md` (2026-11-11), `docs/CRASHRISK.md`, `evidence/g7_report.json`, `evidence/chair_risk.json`, `evidence/demo_twin.json`: G1–G7, A2 metrics, Task 02 and alert-v1 results.
- **[12]** G7 pre-registration and report: 581 matched events, 158 patients, MDE 2.33 mmHg, τ30 +0.13 [−2.63, +3.06], sham +2.91, P6a +0.43, protocol SHA-256 `9ec625ec…c2fd73`.
- **[13]** Task 02 online alert-utility evaluation and alert-v1 rescue-rule sweep: 8.2% at ≤0.25 alerts/session; 14.1% at 0.26 on fold 2; model channel adds no alerts at affordable slices.

External context

- **[1]** Lin et al., "HEMOBP: a benchmarking database from a hospital-based haemodialysis program," *Scientific Data* 2019. DOI 10.1038/s41597-019-0319-8. CC BY 4.0.
- **[4]** Bello et al., "An update on the global disparities in kidney disease burden and care across continents," *Lancet Global Health* 2024; Okpechi et al., "A global assessment of kidney care workforce," *NDT* 2024 (median 11.75 nephrologists pmp, IQR 1.78–24.76).
- **[5]** ISN Global Kidney Health Atlas 2023 (ISN-GKHA): income-group nephrologist density; low-income ~0.35 pmp vs high-income ~29.02 pmp.
- **[6]** Pradhan Mantri National Dialysis Programme (PMNDP), official dashboard: 36 states/UTs, 751 districts, 1,925 centres, 13,824 haemodialysis machines, 33.16 lakh patients. pmndp.mohfw.gov.in.
- **[7]** "Nephrology Services in Uttar Pradesh," *Indian Journal of Nephrology* 2024: 145 nephrologists; only three centres provide dialysis-technician courses; most district-hospital PMNDP dialysis by nurses or technicians with limited working knowledge. indianjnephrol.org.
- **[8]** Stefánsson et al., *Kidney International* 2014 (IDH and mortality); Chang et al., *CJASN* 2011 (IDH and access thrombosis). Quoted via FACTS [2].
- **[9]** Wang et al., *Kidney360* 2023 (refill 30–50% early; refill-to-UF ratio rises before collapse); Mitsides et al. 2019 (plasma refill 0.12–0.36 L/h).
- **[10]** KDOQI 2015 HD adequacy guidelines: UFR >13 mL/kg/h reference ceiling.
- **[11]** Zhang et al., *NDT* 2023: XGBoost, 693 patients, AUROC 0.89 at 15–75 min. We do not claim to beat it; the horizon differs.

The boundary, in full

We can claim: a calibrated pre-anchor nadir forecast; exact fluid arithmetic; a deterministic, versioned review pathway; honest negative results from the twin and oracle tests; a local FHIR export and hash-chained demo audit.

We refuse to claim: validated alert utility, clinical benefit, or a safety layer; a treatment effect from any UF option; prospective, external, symptomatic-IDH, or Indian-unit validation; authentication, role enforcement, immutable audit, or machine integration.

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Data licences: HEMOBP and the Mendeley Dialysis Database are CC BY 4.0. Raw data is not redistributed in this package. No Indian cohort has been collected; deployment in India requires local retraining, recalibration, and ethics approval.